

```
In [9]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import make_regression
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_absolute_error
from sklearn.metrics import mean_squared_error
from sklearn.metrics import r2_score
```

```
In [10]: # Generate dataset

X, y = make_regression(
    n_samples=100,
    n_features=1,
    noise=10,
    random_state=42
)

# Convert into DataFrame

df = pd.DataFrame( X,
    columns=["YearsExperience"]
)

df["Salary"] = y

df.head()
```

```
Out[10]:
```

	YearsExperience	Salary
0	0.931280	50.779929
1	0.087047	-10.065270
2	-1.057711	-34.918392
3	0.314247	10.526743
4	-0.479174	-17.738377

```
In [11]: df.shape
```

```
Out[11]: (100, 2)
```

```
In [12]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 2 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   YearsExperience  100 non-null    float64
1   Salary          100 non-null    float64
dtypes: float64(2)
memory usage: 1.7 KB
```

In [13]: `df.describe()`

Out[13]:

	YearsExperience	Salary
count	100.000000	100.000000
mean	-0.103847	-3.449530
std	0.908168	41.321720
min	-2.619745	-118.027454
25%	-0.600906	-30.170786
50%	-0.126956	-0.634702
75%	0.405952	23.478589
max	1.852278	89.033145

In [14]:

```
plt.figure(figsize=(6,4))

sns.scatterplot(
    x=df["YearsExperience"],
    y=df["Salary"]
)

plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Experience vs Salary")

plt.show()
```

 No description has been provided for this image

In [16]:

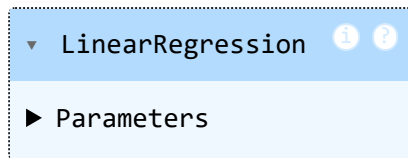
```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

In [17]: `model = LinearRegression()`

In [18]:

```
model.fit(
    X_train,
    y_train
)
```

Out[18]:

In [19]: `y_pred = model.predict(X_test)``y_pred`

```
Out[19]: array([-58.66528327,  65.48743554,  36.04876271, -17.24928234,
                -10.26070235, -12.80652919, -20.50660984, -77.90504757,
                 36.46362633,  41.30294944,  45.71495289,  15.89937588,
                -53.91600662, -23.33958474, -50.82554729, -21.10145005,
                 43.26141852,  10.80464503,  32.7720721 ,  16.72167377])
```

```
In [20]: mae = mean_absolute_error(
          y_test,
          y_pred
        )

        print("MAE:", mae)
```

MAE: 8.41665992220905

```
In [21]: mse = mean_squared_error(
          y_test,
          y_pred
        )

        print("MSE:", mse)
```

MSE: 104.20222653187015

```
In [22]: rmse = np.sqrt(mse)

        print("RMSE:", rmse)
```

RMSE: 10.20794918344866

```
In [23]: print("Coefficient:", model.coef_)
```

Coefficient: [44.24418216]

- Result Interpretation -Linear Regression successfully predicted salary based on years of experience. -The scatter plot shows the relationship between experience and salary. -The regression line represents the predicted trend. -Lower MAE, MSE, and RMSE indicate better prediction accuracy.
- Conclusion The Linear Regression algorithm was successfully implemented without using any external CSV file. The model learned the relationship between Years of Experience and Salary and predicted salary values effectively. Linear Regression is a simple and powerful algorithm for solving regression problems.

In []: